1.1 OSI model: data communication

- The OSI model divides communication into seven layers. Data travels down the sender's stack, crosses the medium, and moves up the receiver's stack. Each layer uses the service below it and serves the layer above.

Sender: functions of the seven layers

- 7. Application: provides network services such as HTTP, email and file transfer. Example: a browser creates an HTTP request.
- 6. Presentation: handles data representation, encoding, compression and encryption. UTF-8 and encryption libraries perform such functions in practical systems.
- 5. Session: establishes, manages and ends a dialogue. It may provide synchronization or checkpoints. Internet applications often implement these functions themselves.
- 4. Transport: provides process-to-process delivery using ports. TCP divides a byte stream into segments and uses sequence numbers, ACKs and retransmission for reliable, ordered delivery.
- 3. Network: adds logical addresses and supports routing between networks. Routers forward IP packets toward the destination using routing information.
- 2. Data link: creates frames for one link, using MAC addresses and an error-detection field such as Ethernet FCS. Switches normally forward frames by destination MAC address.
- 1. Physical: represents bits as electrical, optical or radio signals and carries them through the medium.

Encapsulation and an example

- Encapsulation means adding protocol control information as data moves downward. The data from one layer becomes the payload of the next. Ethernet also adds a trailer.

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Application data
[ TCP header | application data ]
[ IP header | TCP header | application data ]
[ MAC header | IP | TCP | data | FCS ]   ->   bits on the link
```

1.1 OSI model: data communication (cont.)

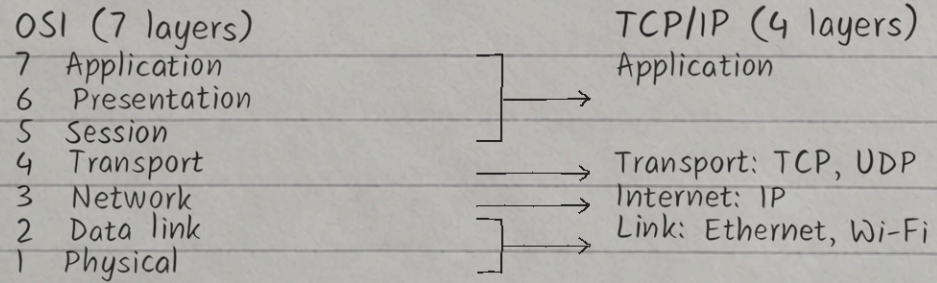
- For an HTTP file request over TCP, TCP adds ports and sequence information; IP adds the endpoint IP addresses; Ethernet adds local-link MAC addresses. The physical interface sends the resulting bits as signals.
- If the server is remote, the first destination MAC belongs to the default gateway. A router removes the incoming frame and creates a new outgoing frame.
- Link addresses change along the route; endpoint IP addresses normally stay the same when there is no NAT.

Receiver: decapsulation

- The receiver recovers bits, checks the Ethernet frame and removes its link header and trailer. IP processes its header and selects TCP. TCP checks, reorders and acknowledges the bytes, removes duplicates, and delivers the stream to the correct application.
- The application interprets the message and displays or stores the file. Thus encapsulation adds control information at the sender, while decapsulation processes and removes it at the receiver.

1.2 OSI and TCP/IP models

- OSI is a seven-layer reference architecture. TCP/IP describes the Internet protocol suite and is commonly shown with four layers. A five-layer teaching version separates physical and data-link functions; the mapping below uses four layers.



1.2 OSI and TCP/IP models (cont.)

Similarities

- Both models divide communication into manageable tasks, use encapsulation, and distinguish applications, transport and routing. Peer entities follow protocols; neighboring layers interact through service interfaces.
- Both support different underlying media. An application can communicate over Ethernet or Wi-Fi without implementing their signaling details. Ports identify application endpoints, IP addresses support inter-network delivery, and MAC addresses serve local links.

Main differences

- Layer count: OSI has seven layers; TCP/IP has four in this convention. TCP/IP combines OSI application, presentation and session functions in its application layer. Its link layer covers the two lowest OSI layers.
- Purpose: OSI supplies a general framework and terminology. TCP/IP is closely associated with protocols used on operational networks. The OSI reference model is distinct from the historical OSI protocol suite.
- Services: OSI explicitly separates services, interfaces and protocols. TCP/IP has broader boundaries; applications often manage their own encoding, encryption and sessions.
- Network layer: OSI can describe connection-oriented or connectionless network services. TCP/IP uses connectionless, best-effort IP. An end-to-end transport such as TCP can add reliability.
- Examples: TCP/IP application protocols include HTTP, DNS and SMTP; transport includes TCP and UDP; the Internet layer includes IP; the link layer includes Ethernet and Wi-Fi access.

Advantages and limitations